

Statistical Methods I: Linear Model

Lec 04: ANOVA and Diagnostics In Simple Linear Regression

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Section 1

Intro

Review: Lecture 01-03

- Lecture 01: LSE:

$$\{\hat{\beta}_0, \hat{\beta}_1\} = \arg \min_{\beta_0, \beta_1} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$$

- The LSEs are

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x}) y_i}{\sum_{i=1}^n (x_i - \bar{x})^2} = \sum_{i=1}^n \frac{(x_i - \bar{x})}{S_{XX}} y_i.$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = \sum_{i=1}^n \left(\frac{1}{n} - \frac{\bar{x}}{S_{XX}} (x_i - \bar{x}) \right) y_i$$

- Lecture 02: CI, and hypothesis testing related to β_1 .
- Lecture 03: Estimation and prediction, [ANOVA](#).
- This lecture (04): [ANOVA](#).

Section 2

Sum of Squares

Sum of Squares Decomposition

- Total Sum of Squares (SST):

$$SST = \sum_{i=1}^n (y_i - \bar{y})^2$$

The total variation in the response variable.

- Regression Sum of Squares (SSR):

$$SSR = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$$

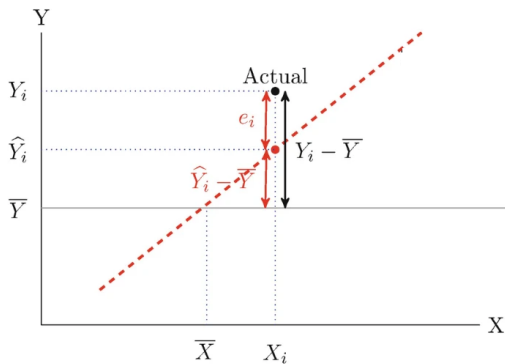
The variation explained by the predictor (regression model).

- Error Sum of Squares (SSE):

$$SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

The variation not explained by the model (regression).

Sums of Squares



Decomposition of the Total Sum of Squares

It can be shown that:

$$SST = SSR + SSE$$

Proof:

$$\begin{aligned} SST &= \sum_{i=1}^n (y_i - \bar{y})^2 \\ &= \sum_{i=1}^n [(y_i - \hat{y}_i) + (\hat{y}_i - \bar{y})]^2 \\ &= \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 + 2 \sum_{i=1}^n (y_i - \hat{y}_i)(\hat{y}_i - \bar{y}) \\ &= SSE + SSR + 2 \sum_{i=1}^n (y_i - \hat{y}_i)(\hat{y}_i - \bar{y}) \end{aligned}$$

The last term is zero because

$$\begin{aligned}\sum_{i=1}^n (y_i - \hat{y}_i)(\hat{y}_i - \bar{y}) &= \sum_{i=1}^n e_i(\hat{y}_i - \bar{y}) \\&= \sum_{i=1}^n e_i(\hat{\beta}_0 + \hat{\beta}_1 X_i - \bar{y}) \\&= \sum_{i=1}^n e_i \hat{y}_i - \bar{y} \sum_{i=1}^n e_i \\&= \sum_{i=1}^n e_i \hat{y}_i - \bar{y} \cdot 0 \quad (\text{since } \sum e_i = 0) \\&= \sum_{i=1}^n e_i(\hat{\beta}_0 + \hat{\beta}_1 X_i) \\&= \hat{\beta}_0 \sum_{i=1}^n e_i + \hat{\beta}_1 \sum_{i=1}^n e_i X_i \\&= \hat{\beta}_0 \cdot 0 + \hat{\beta}_1 \cdot 0 \quad (\text{from normal equations}) \\&= 0.\end{aligned}$$

- Don't remember the normal equations? Here is the derivation

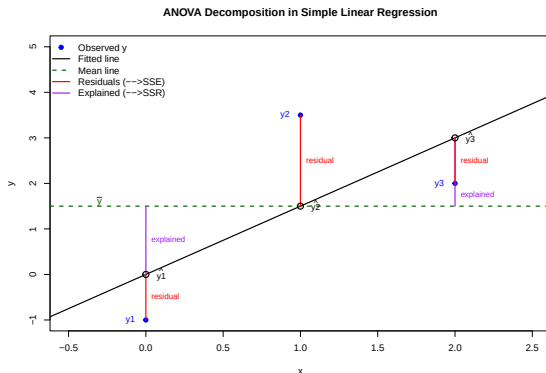
$$\frac{\partial}{\partial \beta_0} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 X_i)^2 = -2 \sum_{i=1}^n (y_i - \beta_0 - \beta_1 X_i) = 0,$$

$$\frac{\partial}{\partial \beta_1} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 X_i)^2 = -2 \sum_{i=1}^n X_i (y_i - \beta_0 - \beta_1 X_i) = 0.$$

- The above two equations are called the normal equations.
They can be rewritten as

$$\sum_{i=1}^n e_i = 0 \quad \text{and} \quad \sum_{i=1}^n e_i X_i = 0.$$

Sum of Squares Decomposition: A Numerical Example



Sum of Squares Decomposition: A Numerical Example

- $(x_1, y_1) = (0, -1); (x_2, y_2) = (1, 3.5); (x_3, y_3) = (2, 2)$
- $\hat{\beta}_0 = 0, \hat{\beta}_1 = 1.5$
- $\bar{y} = \frac{-1+3.5+2}{3} = 1.5$
- $\hat{y}_1 = \hat{\beta}_0 + \hat{\beta}_1 x_1 = 0 + 1.5 * 0 = 0$
- $\hat{y}_2 = \hat{\beta}_0 + \hat{\beta}_1 x_2 = 0 + 1.5 * 1 = 1.5$
- $\hat{y}_3 = \hat{\beta}_0 + \hat{\beta}_1 x_3 = 0 + 1.5 * 2 = 3$
- $SST = \sum_{i=1}^3 (y_i - \bar{y})^2 = (-1 - 1.5)^2 + (3.5 - 1.5)^2 + (2 - 1.5)^2 = 10.5$
- $SSR = \sum_{i=1}^3 (\hat{y}_i - \bar{y})^2 = (0 - 1.5)^2 + (1.5 - 1.5)^2 + (3 - 1.5)^2 = 4.5$
- $SSE = \sum_{i=1}^3 (y_i - \hat{y}_i)^2 = (-1 - 0)^2 + (3.5 - 1.5)^2 + (2 - 3)^2 = 6$

Sum of Squares Decomposition: A Numerical Example

```
# ANOVA components
```

```
SST <- sum((y - ybar)^2)
```

```
SSR <- sum((yhat - ybar)^2)
```

```
SSE <- sum((y - yhat)^2)
```

```
SST; SSR; SSE; SST - (SSR + SSE) # should be 0
```

```
## [1] 10.5
```

```
## [1] 4.5
```

```
## [1] 6
```

```
## [1] 0
```

Section 3

ANOVA Table

ANOVA Table for SLR

- An analysis of variance (ANOVA) table summarizes the results of the regression analysis. It typically has 5 or 6 columns.
 - 1 Source of variation (Regression, Error, Total)
 - 2 Degrees of freedom (DF)
 - 3 Sum of squares (SS)
 - 4 Mean square ($MS = SS/DF$)
 - 5 F-statistic (for testing $H_0 : \beta_1 = 0$ vs. $H_a : \beta_1 \neq 0$)
 - 6 p-value (optional)

ANOVA Table for SLR

Source	DF	SS	MS	F
<i>Regression</i>	1	<i>SSR</i>	$MSR = SSR/1$	$F = MSR/MSE$
<i>Error</i>	$n - 2$	<i>SSE</i>	$MSE = SSE/(n - 2)$	
<i>Total</i>	$n - 1$	<i>SST</i>		

- DF: degrees of freedom
- SS: sum of squares
- MS: mean square (SS divided by corresponding DF)
- F: F-statistic for testing $H_0 : \beta_1 = 0$ vs. $H_a : \beta_1 \neq 0$
- Under H_0 , $F \sim F_{1,n-2}$.
- R^2 can be computed from the table. (how?)

E[SSR]

- We will rewrite SSR in terms of $\hat{\beta}_1$:

$$\begin{aligned} SSR &= \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 \\ &= \sum_{i=1}^n (\hat{\beta}_0 + \hat{\beta}_1 X_i - \bar{y})^2 \\ &= \sum_{i=1}^n (\hat{\beta}_1 (X_i - \bar{X}))^2 \quad (\text{since } \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{X}) \\ &= \hat{\beta}_1^2 \sum_{i=1}^n (X_i - \bar{X})^2 = \hat{\beta}_1^2 S_{XX}. \end{aligned}$$

- Therefore,

$$E[SSR] = E[\hat{\beta}_1^2] S_{XX}.$$

E[MSR]

- We find that $SSR = \hat{\beta}_1^2 S_{XX}$. But $\hat{\beta}_1 \sim N(\beta_1, \sigma^2/S_{XX})$. Note that $Var(\hat{\beta}_1) = E(\hat{\beta}_1^2) - E^2[\hat{\beta}_1]$. Therefore,

$$\begin{aligned} E[SSR] &= E[\hat{\beta}_1^2 S_{XX}] \\ &= S_{XX} E[\hat{\beta}_1^2] \\ &= S_{XX} \left(Var(\hat{\beta}_1) + (E[\hat{\beta}_1])^2 \right) \\ &= S_{XX} \left(\frac{\sigma^2}{S_{XX}} + \beta_1^2 \right) \\ &= \sigma^2 + \beta_1^2 S_{XX}. \end{aligned}$$

Note that $MSR = SSR/1 = \hat{\beta}_1^2 S_{XX}$, and
 $E[MSR] = E[SSR/1] = \sigma^2 + \beta_1^2 S_{XX}$.

E[SSE]

$$\begin{aligned} E[SSE] &= E \left[\sum_{i=1}^n (y_i - \hat{y}_i)^2 \right] = E \left[\sum_{i=1}^n e_i^2 \right] = E[e'e] \\ &= E[(Y - \hat{Y})'(Y - \hat{Y})] = E[Y'(I - H)'(I - H)Y] \\ &= E[Y'(I - H)Y] \quad (\text{since } I - H \text{ is symmetric and idempotent}) \\ &= E[\text{trace}((I - H)YY')] \quad (\text{since trace is invariant under cyclic permutation}) \\ &= \text{trace}((I - H)E[YY']) \\ &= \text{trace}((I - H)(\text{Var}(Y) + E[Y]E[Y'])) \\ &= \text{trace}((I - H)(\sigma^2 I + X\beta\beta'X')) \\ &= \sigma^2 \text{trace}(I - H) + \text{trace}((I - H)X\beta\beta'X') \\ &= \sigma^2(n - 2) + 0 \quad (\text{since } (I - H)X = 0) \\ &= (n - 2)\sigma^2. \end{aligned}$$

E[MSE]

- Because $MSE = SSE/(n - 2)$, we have

$$E(MSE) = \sigma^2.$$

- When $\beta_1 = 0$, $E(MSR) = E(MSE) = \sigma^2$.
- When $\beta_1 \neq 0$, $E(MSR) > E(MSE)$.
- This justifies the F-test for testing $H_0 : \beta_1 = 0$ vs. $H_a : \beta_1 \neq 0$.

Section 4

F Test

Construct F Test Statistics

- F-distributed statistics can be constructed as the ratio of two independent chi-square distributed statistics divided by their degrees of freedom. Let
 - $U \sim \chi_{d_1}^2$
 - $V \sim \chi_{d_2}^2$
 - U and V are independent.
- Then

$$F = \frac{U/d_1}{V/d_2} \sim F_{d_1, d_2}.$$

F-test in SLR

- In SLR, we can show that
 - $SSE/\sigma^2 \sim \chi_{n-2}^2$
 - $SSR/\sigma^2 \stackrel{\text{when?}}{\sim} \chi_1^2$
 - SSR and SSE are independent.
- Therefore,

$$F = \frac{MSR}{MSE} = \frac{SSR/1}{SSE/(n-2)} \stackrel{\text{when?}}{\sim} F_{1,n-2}.$$

F Test

- It turns out that

$$SSR/\sigma^2 \stackrel{H_0: \beta_1=0}{\sim} \chi_1^2$$

- Also, we have shown that

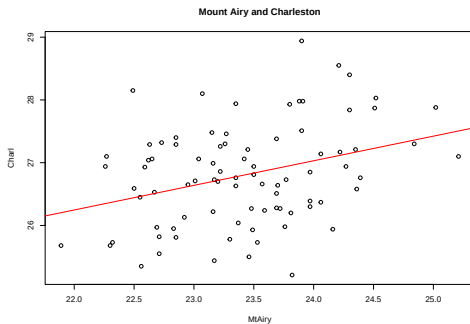
$$E[MSR] = \sigma^2 + \beta_1^2 S_{XX}, E[MSE] = \sigma^2$$

- Both suggest that we should reject $H_0 : \beta_1 = 0$ when the ratio $F = MSR/MSE$ is too large.
- We reject H_0 if F is too large: the rejection region is $F > F_{\alpha,1,n-2}$, where $F_{\alpha,1,n-2}$ is the $(1 - \alpha)$ quantile of the $F_{1,n-2}$ distribution.
- The p-value is $P(F_{1,n-2} \geq F_{\text{obs}})$, where F_{obs} is the observed value of the F-statistic.

Example

Summer temperature averages in Mount Airy, NC and Charleston, SC (1937-2021)

```
mtair=read.csv("data/mta1.csv", head=TRUE)
plot(mtair$MtAiry, mtair$Charl,
     xlab="MtAiry",
     ylab="Charl",
     main="Mount Airy and Charleston")
fit <- lm(Charl ~ MtAiry, data = mtair)
abline(fit, col="red", lwd=2)
```



Example: ANOVA Table

```
anova(fit)
```

```
## Analysis of Variance Table
##
## Response: Charl
##      Df Sum Sq Mean Sq F value    Pr(>F)
## MtAiry    1  6.047   6.0470  10.521 0.001701 **
## Residuals 83 47.705   0.5748
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

F test and t-test

- The F-statistic can be rewritten as

$$F = \frac{MSR}{MSE} = \frac{\hat{\beta}_1^2 S_{XX}}{s^2} = \left(\frac{\hat{\beta}_1}{s/\sqrt{S_{XX}}} \right)^2 = t^2,$$

where $t = \hat{\beta}_1/(s/\sqrt{S_{XX}})$ is the t-statistic for testing $H_0 : \beta_1 = 0$.

- Thus, the F-test and the t-test for testing $H_0 : \beta_1 = 0$ are equivalent in SLR.
- In fact, as $F_{1,n-2}$ distribution is also equivalent to the distribution of the square of a random variable whose distribution is t_{n-2} , the F test is equivalent to the two-sided t test.

Example

- Find the p-values based on t- and F-tests. What can you say about them?

```
summary(fit)
```

```
##
## Call:
## lm(formula = Charl ~ MtAiry, data = mtair)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.7502 -0.6291 -0.0123  0.5585  1.9484
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  17.6208     2.8325   6.221 1.91e-08 ***
## MtAiry        0.3921     0.1209   3.244  0.0017 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7581 on 83 degrees of freedom
## Multiple R-squared:  0.1125, Adjusted R-squared:  0.1018
## F-statistic: 10.52 on 1 and 83 DF, p-value: 0.001701
```

Section 5

F test to Compare Models

Compare Two Nested Models

- F tests can be used to compared nested models
 - Here, nested models mean that one model (the reduced model) is a special case of the other model (the full model) with some parameters set to zero.
- For Simple Linear Regression (SLR), the two following models are nested:
 - Reduced model: $y_i = \beta_0 + \epsilon_i$, $\rightarrow SSE_R$, note $SSE_R = SST$.
 - Full model: $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$, $\rightarrow SSE_F$, note $SSE_F = SSE$.
- The reduced model is obtained from the full model by setting $\beta_1 = 0$.
- The difference in SEE between the two models is due to the inclusion of the predictor variable x in the full model:
$$SSE_R - SSE_F = SST - SSE = SSR.$$

Compare Two Nested Models

- Let SSE_F and SSE_R be the SSE for the full and reduced models, respectively.
- Note that $SSE_F \leq SSE_R$ because the full model has more parameters and thus fits the data at least as well as the reduced model.
- We can construct the following F-statistic to compare the two models:

$$F = \frac{(SSE_R - SSE_F)/(df_R - df_F)}{SSE_F/df_F}$$

where $df_F = n - 2$ and $df_R = n - 1$ are the degrees of freedom for the full and reduced models, respectively.

- Under $H_0 : \beta_1 = 0$, $F \sim F_{df_F - df_R, df_F}$.

Example: Compare Two Nested Models

```
# Reduced model:  $y_i = \beta_0 + \epsilon_i$ 
fit_reduced <- lm(Charl ~ 1, data = mtair)
# Full model:  $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$ 
fit_full <- lm(Charl ~ MtAiry, data = mtair)
anova(fit_reduced, fit_full)
```

```
## Analysis of Variance Table
```

```
##
```

```
## Model 1: Charl ~ 1
```

```
## Model 2: Charl ~ MtAiry
```

```
##   Res.Df    RSS Df Sum of Sq      F   Pr(>F)
```

```
## 1      84 53.752
```

```
## 2      83 47.705   1     6.047 10.521 0.001701 **
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Section 6

F test and R^2

F test and R^2

- The coefficient of determination, R^2 , is defined as

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}.$$

- R^2 measures the proportion of the variance in the response variable that is explained by the predictor variable.
- R^2 is always between 0 and 1. A value of 0 indicates that the model does not explain any of the variance in the response variable, while a value of 1 indicates that the model explains all of the variance.

F test and R^2

- The F-statistic can be rewritten in terms of R^2 as follows:

$$\begin{aligned} F &= \frac{MSR}{MSE} = \frac{SSR/1}{SSE/(n-2)} \\ &= \frac{(R^2 \cdot SST)/1}{((1 - R^2) \cdot SST)/(n-2)} \\ &= \frac{R^2}{1 - R^2} \cdot (n-2). \end{aligned}$$

- Therefore, the F-test for testing $H_0 : \beta_1 = 0$ vs. $H_a : \beta_1 \neq 0$ is equivalent to testing whether R^2 is significantly greater than 0.

F test and R^2

- The p-value for the F-test can be computed as

$$\text{p-value} = P(F_{1,n-2} \geq F_{\text{obs}}) = P\left(F_{1,n-2} \geq \frac{R^2}{1-R^2} \cdot (n-2)\right).$$

- Because of the relationship between F and R^2 , under $H_0 : \beta_1 = 0$, we have

$$\frac{R^2}{1-R^2} \cdot (n-2) \stackrel{H_0: \beta_1=0}{\sim} F_{1,n-2}.$$

- Is there a name of the null distribution of R^2 ?

F test and R^2

- Yes, its called the Beta distribution. Specifically, under $H_0 : \beta_1 = 0$,

$$R^2 \stackrel{H_0: \beta_1=0}{\sim} \text{Beta} \left(\frac{1}{2}, \frac{n-2}{2} \right).$$

- In fact, there is a connection between F and Beta distributions:
- If $F \sim F_{d_1, d_2}$, then $\frac{d_1 F}{d_1 F + d_2} \sim \text{Beta} \left(\frac{d_1}{2}, \frac{d_2}{2} \right)$.
- Conversely, if $B \sim \text{Beta} \left(\frac{d_1}{2}, \frac{d_2}{2} \right)$, then $\frac{d_2 B}{d_1(1-B)} \sim F_{d_1, d_2}$.

Correlation Coefficient

- The correlation coefficient, r , between the response variable y and the predictor variable x is defined as

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2}}.$$

- We have shown the relationship between $\hat{\beta}_1$ and r :

$$\hat{\beta}_1 = r \frac{S_Y}{S_X},$$

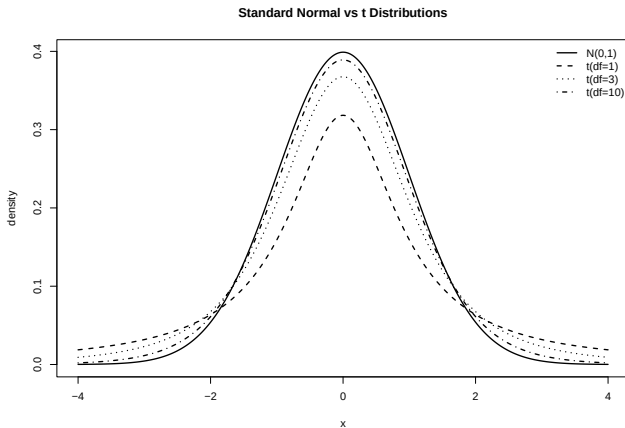
where S_Y and S_X are the sample standard deviations of y and x , respectively.

- It can be shown that:

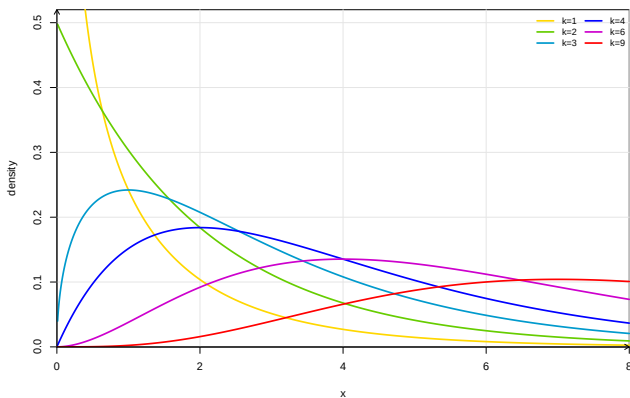
$$t = r \sqrt{\frac{n-2}{1-r^2}}, \text{ which has been shown } \stackrel{H_0: \beta_1=0}{\sim} t_{n-2}.$$

Section 7

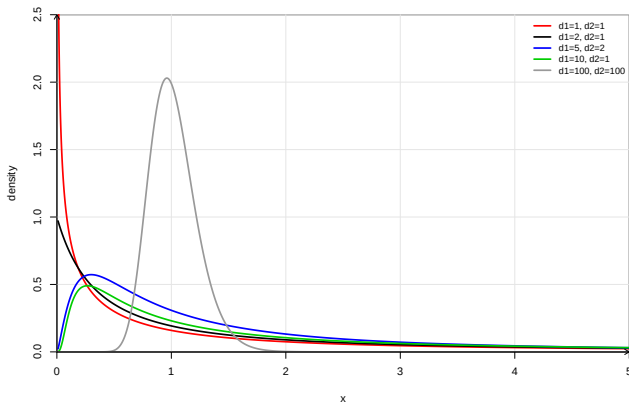
Distributions

t vs $N(0,1)$ 

Chi-square



F distribution



Beta Distribution

